

-
- Yifan Song, Guoyin Wang, Sujian Li, and Bill Yuchen Lin. The good, the bad, and the greedy: Evaluation of llms should not ignore non-determinism. *arXiv preprint arXiv:2407.10457*, 2024.
- Taylor Sorensen, Jared Moore, Jillian Fisher, Mitchell Gordon, Niloofar Mireshghallah, Christopher Michael Rytting, Andre Ye, Liwei Jiang, Ximing Lu, Nouha Dziri, et al. Position: a roadmap to pluralistic alignment. In *Proceedings of the 41st International Conference on Machine Learning*, pp. 46280–46302, 2024.
- Guy Tevet and Jonathan Berant. Evaluating the evaluation of diversity in natural language generation. In *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: Main Volume*, pp. 326–346, 2021.
- Yufei Tian, Tenghao Huang, Miri Liu, Derek Jiang, Alexander Spangher, Muhao Chen, Jonathan May, and Nanyun Peng. Are large language models capable of generating human-level narratives? In *EMNLP*, 2024.
- Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.
- Saranya Venkatraman, Nafis Irtiza Tripto, and Dongwon Lee. Collabstory: Multi-llm collaborative story generation and authorship analysis. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pp. 3665–3679, 2025.
- Vivek Verma, Nicholas Tomlin, and Dan Klein. Revisiting entropy rate constancy in text. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 15537–15549, 2023.
- Ziyu Wan, Xidong Feng, Muning Wen, Stephen Marcus McAleer, Ying Wen, Weinan Zhang, and Jun Wang. Alphazero-like tree-search can guide large language model decoding and training. In *Forty-first International Conference on Machine Learning*, 2024.
- Chaoqi Wang, Yibo Jiang, Chenghao Yang, Han Liu, and Yuxin Chen. Beyond reverse kl: Generalizing direct preference optimization with diverse divergence constraints. In *The Twelfth International Conference on Learning Representations*, 2024.
- Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin Yang, Zhenru Zhang, et al. Beyond the 80/20 rule: High-entropy minority tokens drive effective reinforcement learning for llm reasoning. *arXiv preprint arXiv:2506.01939*, 2025.
- Yuxia Wang, Daniel Beck, Timothy Baldwin, and Karin Verspoor. Uncertainty estimation and reduction of pre-trained models for text regression. *Transactions of the Association for Computational Linguistics*, 10: 680–696, 2022.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS*, 2022.
- Sean Welleck, Amanda Bertsch, Matthew Finlayson, Hailey Schoelkopf, Alex Xie, Graham Neubig, Ilya Kulikov, and Zaid Harchaoui. From decoding to meta-generation: Inference-time algorithms for large language models. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=eskQMCIbMS>. Survey Certification.
- Peter West and Christopher Potts. Base models beat aligned models at randomness and creativity. *arXiv preprint arXiv:2505.00047*, 2025.
- Fan Wu, Emily Black, and Varun Chandrasekaran. Generative monoculture in large language models. In *The Thirteenth International Conference on Learning Representations*, 2025a. URL <https://openreview.net/forum?id=yZ7sn9pyqb>.
- Fang Wu, Weihao Xuan, Ximing Lu, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why rlvr may not escape its origin. *arXiv preprint arXiv:2507.14843*, 2025b.

-
- Wilson Wu, John X Morris, and Lionel Levine. Do language models plan ahead for future tokens? In *The First Conference on Language Modeling*, 2024.
- Miao Xiong, Zhiyuan Hu, Xinyang Lu, YIFEI LI, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=gjeQKFxFpZ>.
- An Yang, Baosong Yang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Zhou, Chengpeng Li, Chengyuan Li, Dayiheng Liu, Fei Huang, et al. Qwen2 technical report. *arXiv preprint arXiv:2407.10671*, 2024.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Tom Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. *Advances in neural information processing systems*, 36:11809–11822, 2023.
- Fanghua Ye, Mingming Yang, Jianhui Pang, Longyue Wang, Derek F Wong, Emine Yilmaz, Shuming Shi, and Zhaopeng Tu. Benchmarking llms via uncertainty quantification. *arXiv preprint arXiv:2401.12794*, 2024.
- Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, et al. Lima: Less is more for alignment. *Advances in Neural Information Processing Systems*, 36, 2024.

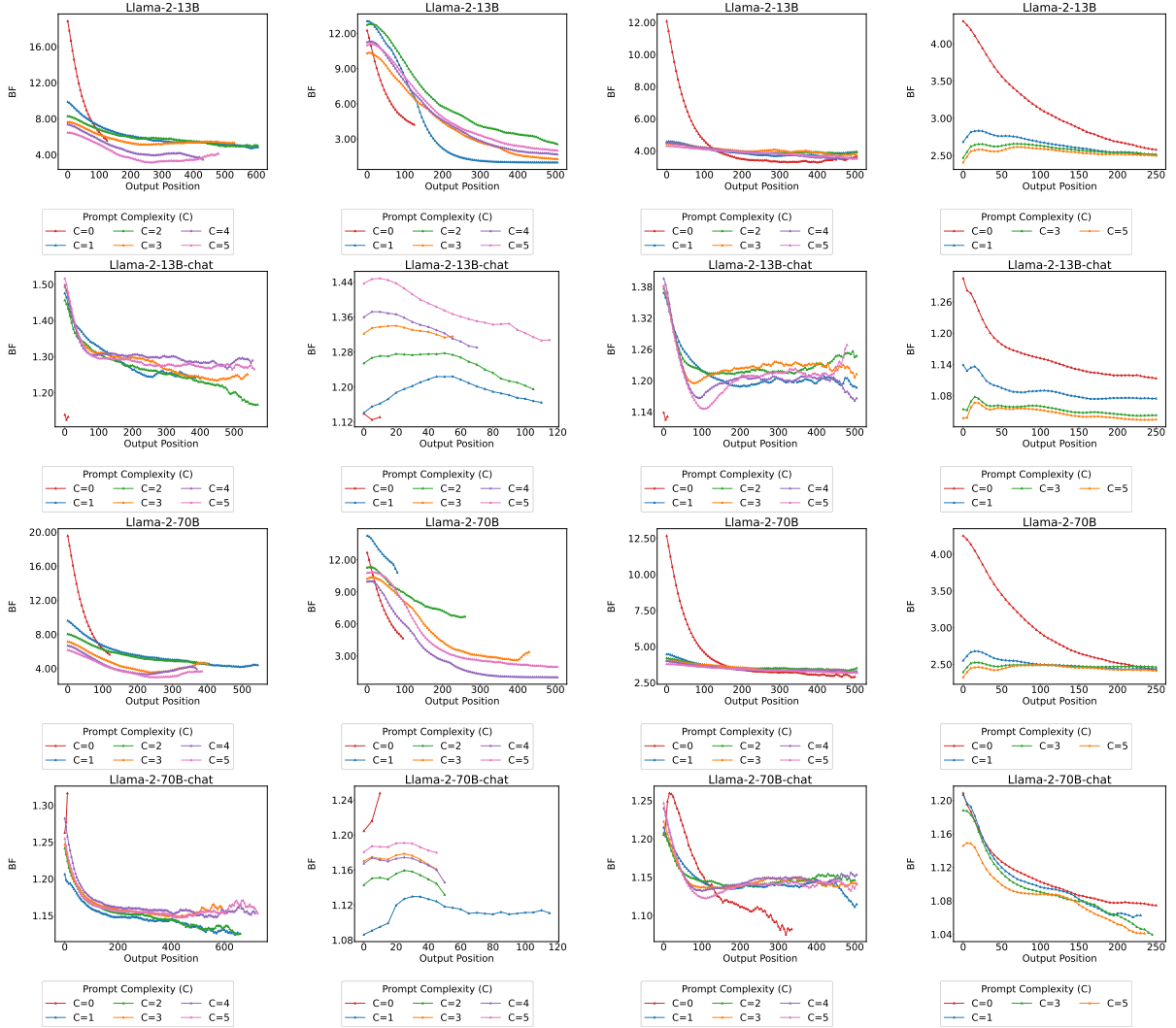


Figure 8: **BF Output Dynamic for Llama-2-families.** For better visualization, we compute the exponential moving averaged values of perplexity with the smoothing factor set as 0.1.

A Case Study Implementation Details

We use the scripts in Qwen-2.5-Math (Yang et al., 2024) for standard reasoning benchmarks.¹⁰ We sample 200 examples from MMLU-STEM and compute the performance numbers under 64 trials and report the average performance.

B Dataset-Specific Processing

For all datasets we used in the paper, we carefully controlled whether the prompt length and expected output length would exceed the model’s maximum length.

MMLU (Hendrycks et al., 2021) is a widely-used multiple-choice reasoning question. Unless otherwise explained, we use the full test set of MMLU to avoid potential contamination, following benchmarking settings reported in most LLM technical reports (Touvron et al., 2023; Dubey et al., 2024; Guo et al., 2025).

¹⁰<https://github.com/QwenLM/Qwen2.5-Math/tree/main>